

Review

Survivable Elastic Optical Network: A survey of failure scenarios and solutions

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ABSTRACT

Survivable Elastic Optical Networks (SEON) are at the forefront of high-speed, flexible, and bandwidth-efficient optical communication. Modern flex-grid Elastic Optical Networks (EON) allocate variable-width frequency slots to match heterogeneous traffic demands and modulation formats; the survivability mechanisms discussed here operate at the spectrum slot layer and are agnostic to the underlying transponder waveform. Despite these advantages, ensuring survivability against diverse failures, including single-link, multi-link, node and hardware outages, remains a major challenge. This paper presents a comprehensive survey of SEON survivability, with a particular emphasis on classifying solutions according to different failure scenarios. State-of-the-art strategies such as Dedicated Path Protection, Shared Backup Path Protection, segmented and span restoration, and p-cycle protection are systematically analyzed and compared. Unlike previous surveys, this work integrates a classification framework with performance insights, highlighting how existing mechanisms fare under different types of failures. To bridge existing gaps, we also discuss open research challenges, including centralized control, spectrum management, AI-driven resilience, physical layer impairments and disaster recovery. Furthermore, we outline the potential of a hybrid protection paradigm combining state-aware traffic splitting, dynamic allocation, and vulnerability scoring to improve spectrum utilization and reduce blocking probability. By linking failure scenarios with recovery strategies and future research needs, this survey provides a unique perspective to guide the design of scalable, adaptive and intelligent SEON architectures.

Contents

1. Introduction	2
2. State of the art of SEON.....	2
2.1. Dedicated path and shared backup path protection scheme.....	4
2.2. Segmented protection and restoration schemes	5
2.3. Path and span restoration schemes	6
2.4. Ring cover and p-Cycle protection schemes	6
3. Different failures scenarios	8
3.1. Single and dual-link/multi-link failure scenarios	8
3.2. Node failure.....	8
3.3. Path failures	8
3.4. Hardware failures.....	9
4. Open challenges and future researches	10
4.1. Centralized network control and management	10
4.2. Spectrum management.....	12
4.3. Artificial intelligence in SEON	12
4.4. Physical layer impairments.....	13
4.5. Hardware development.....	14
4.6. Disaster management.....	15
5. Implementation and simulation results	15
6. Conclusion	19

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CRediT authorship contribution statement	19
Declaration of competing interest	19
Acknowledgments	19
Data availability	19
References	19

1. Introduction

The technology associated with Elastic Optical Networks, particularly in terms of their resilience and adaptability, has undergone significant advancements in recent years. With the rapid evolution of global internet connectivity encompassing 5G and beyond, WiFi, the Internet of Things (IoT), mobile technology [1], machine-to-machine (M2M) communication [2], and centralized control paradigms such as Software-Defined Networking (SDN) [3], the global digital landscape is set to transform dramatically. According to Cisco's Annual Internet Report (2018–2023) projected that by 2023 roughly two-thirds of the world's population (about 5.3 billion people) would be online [4]. Recent ITU estimates are consistent with that trajectory: around 5.5 billion people about 68% of the global population were online in 2024, with roughly 2.6 billion still offline [5]. A comprehensive list of the acronyms and abbreviations used throughout this paper is provided in Table 1 for ease of reference.

Simultaneously, the field of optical networks has experienced rapid development, evolving from TON [6] based on WDM and fixed-grid technology. Given that networking resources are both crucial and finite, inefficient utilization of spectrum resources becomes an even greater concern, particularly as frequency spacing expands. A significant portion of the available spectrum can remain underutilized when channels are assigned fixed bandwidths, creating large unused frequency gaps where no traffic can be transmitted. This inefficiency is inherent in WDM systems due to their coarse granularity, with fixed 50 GHz or 100 GHz channel spacing. As a result, traditional WDM network architectures [7] exhibit limited elasticity and flexibility at the optical layer.

The demand for highly dynamic and adaptive traffic management is becoming increasingly critical in today's connectivity-driven world [8, 9]. To address the rigid constraints of TONs, researchers have proposed spectrum-efficient EON built on Orthogonal Frequency-Division Multiplexing (OFDM) [10–12]. Due to their ability to dynamically allocate large optical bandwidths based on heterogeneous traffic demands, flexi-grid EON are widely recognized as a promising solution for future ultra-high-speed optical transmission systems [13].

One of the first EON architectures, SLICE, was introduced in [14]. Compared to traditional WDM networks, this EON model features finer spectrum granularity, allocating an appropriate number of FSs for each optical channel based on the actual data rate and modulation format. Unlike conventional 50 GHz or 100 GHz fixed-grid frequency allocation, EON align optical channels to a narrower grid frequency, enabling greater flexibility in bandwidth allocation and improving spectrum utilization efficiency.

However, spectrum continuity constraints impose a significant challenge in EON. Each optical channel must utilize the same set of contiguous FSs across all fiber links it traverses, which complicates network management. While a similar limitation, wavelength continuity exists in traditional WDM networks, the spectrum continuity constraint in EON is more complex because it requires ensuring the continuity of multiple FSs across different links.

The EON has drawn a lot of interest from both academia and business since it was initially put forth. Numerous relevant studies have been conducted, including:

- Distributed network control and management based on the GMPLS [41–48].
- Traffic grooming such as bandwidth squeezing [49,50].
- RSA [51].
- Issues regarding survivability, such as single and multi-link failures on EON [8].

Since all of the previously mentioned studies concerning the EON have been skimmed over by researchers working in fields like GMPLS, RSA and network control and management. Network survivability is one of the issues that has to be further investigated in terms of various unique constraints and specific primary techniques for restoration and protection that aid our researchers in the field of SEON development.

The novelty of this survey lies in its comprehensive treatment of survivability in SEON by unifying various failure scenarios ranging from single link to hardware and node-level disruptions into a single structured survey framework. Unlike existing surveys that focus narrowly on either protection or restoration, our work systematically examines both strategies under the lens of modern SEON requirements. Furthermore, we introduce a hybrid protection perspective that integrates symmetrical traffic splitting, dynamic allocation and a vulnerability scoring. This integrated model not only enhances resource efficiency and blocking probability performance but also provides a fresh direction for intelligent and adaptive SEON design.

The main contributions of this paper can be summarized as follows:

- We present a detailed classification of failure scenarios in SEON, covering single-link, multi-link, node and hardware failures and analyze their implications on spectrum allocation and survivability.
- We survey state-of-the-art survivability mechanisms and benchmark their effectiveness in SEON.
- We propose a hybrid protection model that combines Vulnerability scoring-based path selection with traffic splitting, thereby reducing blocking probability and improving spectrum utilization.
- We identify open research challenges such as centralized control integration, AI-driven resilience and disaster recovery, outlining promising directions for future work.

This paper begins by presenting a comprehensive overview of the existing state of the art in the field of SEON. Subsequently, it delves into a discourse on recent progress and scholarly investigations pertaining to various schemes in Section 2. Finally, an examination of diverse failure scenarios is conducted with the objective of ensuring the preservation of survivability in Section 3. In light of the literature review and investigations discussed before, we now proceed to examine the future research challenges pertaining to the SEON in Section 4. The work's conclusion is given in Section 5.

2. State of the art of SEON

This section discusses and reviews the state-of-the-art techniques for ensuring survivability in EON, as illustrated in Fig. 1. The focus is on the most widely adopted protection and restoration schemes, which play a critical role in maintaining network resilience against failures. EON are capable of supporting individual data rates ranging from 400 Gbps to 1000 Gbps [52], with throughput per fiber link reaching approximately 10–100 Tbps. Given these high transmission capacities, a failure in a network node or optical fiber can lead to significant data

- Centralized network control and management such as SDON, SDEON [12,15–32,32–40].

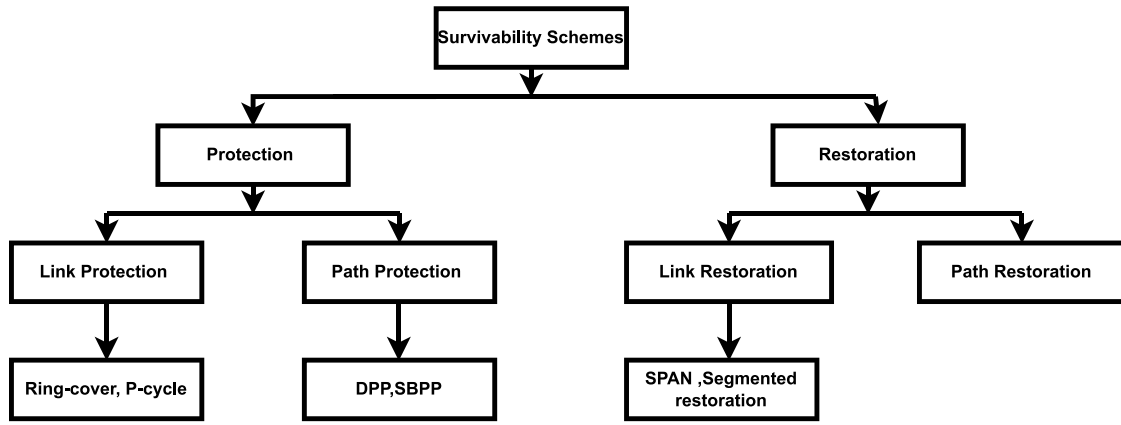


Fig. 1. Different protection and restoration schemes.

Table 1

List of acronyms used in this paper.

Acronym	Definition
AI	Artificial Intelligence
BVT	Bandwidth Variable Transponder
BLSR	Bidirectional Line Switched Ring
BER	bit error ratio
CAPEX	Capital expenditures
CON	Content Oriented Networking
DES	Double Exponential Smoothing
DPP	Dedicated Path Protection
DRL	Deep Reinforcement Learning
EON	Elastic Optical Network
EDFA	Erbium-Doped Fiber Amplifier
EGA	Enhanced Genetic Algorithm
FS	Frequency Slot
FIPP	Failure Independent Path Protecting <i>p</i> -cycle
GMPLS	Generalized Multiprotocol Label Switching
GPCPS	Genetic <i>p</i> -cycle Combination Protection Technique
MSALPF	Subcarriers and Average Longest Path First
MPLS	Multi Protocol Label Switching
NWDM	Nyquist Wavelength Division Multiplexing
NN	Neural Network
OFDM	Orthogonal Frequency Division Multiplexing
OSNR	Optical Signal-To-Noise Ratios
PLI	Physical Layer Impairments
PDPP	Partitioned Dedicated Path Protection
PR	Path Restoration
QoT	Quality-Of-Transmission
QoS	Quality of Service
QPSK	Quadrature Phase Shift Keying
RSA	Routing and Spectrum Assignment
RMSA	Routing, Modulation and Spectrum Assignments
ROADM	Reconfigurable optical add-drop multiplexer
SDN	Software Defined Networking
SDON	Software Defined Optical Network
SD-SEON	Software Defined survivable elastic optical network
SEON	Survivable Elastic Optical Network
SVM	Support Vector Machine
SR	Span Restoration
SBPP	Shared Backup Path Protection
SFR	Spectrum Fragmentation Ratio
SBVT	Sliceable Bandwidth Variable Transponder
S-RMLSA	Survivable Routing, Modulation Level and Spectrum Assignment
SDM	Space Division Multiplexed
TON	Traditional Optical Networks
UPSR	Unidirectional Path Switched Ring
WDM	Wavelength Division Multiplexing
WSS	Wavelength Selective Switch

loss, disruption of services for millions of users, and substantial financial losses. Consequently, ensuring network survivability has become

a fundamental requirement for EON [53]. The core mechanisms for ensuring survivability are protection and restoration strategies, which are further classified into: Link Protection and Restoration, and Path Protection and Restoration, as depicted in Fig. 1. These strategies aim to minimize service disruptions and enhance network robustness by preemptively securing backup resources (protection) or dynamically recovering from failures (restoration).

Protection-based techniques in EON are designed to ensure network resilience by preemptively allocating backup resources or paths to mitigate potential failures. These methods are broadly categorized into Ring-Cover, *p*-Cycle, DPP, and SBPP. The core principle of protection-based survivability is to establish fail-safe routes in advance, which remain idle until a failure occurs. When a disruption is detected, the affected traffic is swiftly rerouted to a pre-configured backup path, enabling fast recovery with minimal service interruption. This proactive approach significantly enhances network reliability by reducing recovery time compared to reactive methods. Additionally, protection-based techniques offer efficient resource allocation, as some methods, such as SBPP, allow for resource sharing among multiple connections, optimizing bandwidth utilization. These techniques are highly scalable and adaptable, ensuring robust protection at both the link and path levels. By employing these strategies, EON can effectively handle network failures while maintaining high availability, seamless connectivity, and minimal data loss. The following are the primary traits of protection-based survivability.

- **Fast Failover:** Because the backup link is already built and available to be used when necessary, protection-based solutions provide fast failover speeds.
- **Paths already established:** Prior to any failure, backup routes are calculated and set up. This means that even if they are not being used right now, the network must commit resources for protection.
- **More Resource Overhead:** Compared to restoration-based procedures, protection-based strategies may have more resource overhead since they need additional network resources to support the backup channels.

Unlike protection-based methods, which rely on pre-configured backup paths, restoration-based solutions focus on dynamically identifying alternative routes in real-time when a failure occurs. These techniques leverage the network management system to detect failures and compute new paths as needed, rather than reserving dedicated backup resources in advance. This dynamic approach provides greater resource efficiency since spare capacity is not pre-allocated, making it more adaptable to unpredictable network conditions. However, restoration-based methods typically involve longer recovery times compared to

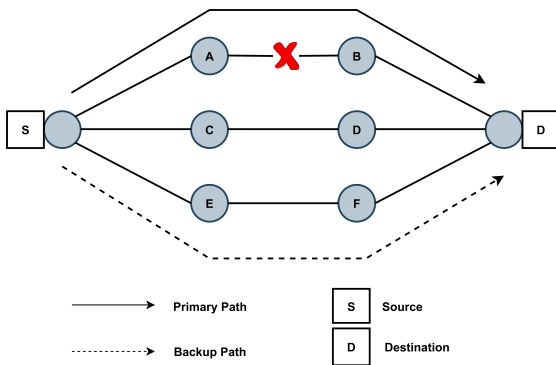


Fig. 2. Dedicated Path Protection Scheme.

protection-based techniques, as new paths must be calculated and established after a failure is detected. Despite this drawback, restoration techniques are highly flexible and scalable, allowing networks to efficiently adapt to varying traffic demands and failure scenarios. By employing intelligent routing algorithms and real-time network monitoring, restoration-based survivability enhances fault tolerance while optimizing resource utilization in EON. The following are the primary traits of restoration-based survivability:

- **Real-Time Path Calculation:** When a failure is recognized, restoration-based techniques use real-time path calculation. Compared to protection-based systems, which have pre-established pathways, this could cause some delay.
- **Utilization of On-Demand Resources:** Until a failure takes place, restoration-based solutions do not use up extra network resources. The resources are employed when needed, which may result in a higher level of resource utilization efficiency.
- **Flexibility and Adaptability:** Restoration-based techniques offer more flexibility to handle various failure scenarios and adapt to changes in network conditions.

In SEON, restoration-based solutions focus on recovering disrupted traffic and services following a network failure. These methods swiftly detect alternative channels and reroute affected traffic to ensure minimal service disruption. Unlike protection-based techniques, which rely on pre-configured backup paths, restoration methods dynamically identify and allocate available resources in real time. Common restoration strategies in SEON include local restoration, end-to-end restoration, segment-based restoration, and dynamic restoration, each offering varying levels of recovery speed and resource efficiency.

Network protection and restoration techniques are generally classified into link-based and path-based approaches. In link-based restoration, failures caused by a damaged or failed link are addressed by rerouting traffic to a pre-established backup path (a set of links). This method enables a rapid localized response and minimizes network downtime. On the other hand, path-based restoration focuses on protecting or restoring entire communication paths (lightpaths) rather than individual links. Since a communication path consists of a series of interconnected nodes and links connecting the source and destination, path-based solutions involve the network management system dynamically determining an alternate route to bypass the affected area. This approach ensures that traffic is successfully rerouted to its intended destination, even in the case of major failures. Path-based solutions primarily rely on the two end nodes of the communication path to coordinate the recovery process. While link-based restoration offers faster failure recovery, path-based restoration provides a more comprehensive and flexible fault-tolerant mechanism, making both approaches essential for ensuring robust network survivability in SEON.

2.1. Dedicated path and shared backup path protection scheme

Among various network survivability techniques, DPP is highly recommended due to its short recovery time and ability to pre-provision backup network resources, such as lightpaths, for each connection. In EON, network resources refer to the spectrum resources of optical fiber links, and path protection is equivalent to allocating backup lightpaths for functional lightpath connections [54].

To illustrate, consider a scenario where a virtual link transmits 100 Gbit/s of traffic. One way to protect this traffic against failures is by establishing an additional physical path with the same 100 Gbit/s capacity, but along a separate route, as shown in Fig. 2. If a failure occurs on the primary path S-A-B-D, for example, the failure of link A-B—the affected virtual link is restored using the backup path S-C-D. This methodology is referred to as DPP [55].

DPP can be categorized into three types: 1+1 protection, 1:1 protection, and quasi 1+1 protection. In 1+1 protection, traffic is simultaneously transmitted on both the working and backup paths, ensuring instantaneous recovery in the event of failure. Switching equipment is deployed at the receiver, which continuously monitors both paths and instantly switches to the backup path if a failure occurs. While this method ensures rapid recovery, it requires additional hardware and dedicated spectrum allocation. Conversely, in 1:1 protection, traffic is transmitted only on the working path under normal conditions, and only switched to the backup path upon failure detection. While this approach reduces resource consumption, it requires longer recovery time compared to 1+1 protection. However, in 1:1 protection, the backup path can be temporarily utilized for lower-priority traffic that would otherwise be discarded in case of a failure [56]. A hybrid approach, quasi 1+1 protection, attempts to balance dedicated and shared protection by grouping multiple critical paths with similar protection requirements. This technique optimizes spectrum allocation and resource utilization, improving network efficiency [57–60].

A defragmentation-based approach using 1+1 path protection in EON has been explored in [57–60]. The primary objective is to enhance network traffic acceptance rates by optimizing the spectrum allocation for backup paths. This method enables dynamic rerouting and spectrum reallocation, allowing primary and backup paths to exchange roles based on varying network conditions. Another strategy, Partitioned DPP (PDPP), was introduced in [55,61,62], where the total transmission rate is divided into multiple disjoint paths. These paths may carry equal or unequal portions of the traffic, thereby reducing spectral usage and minimizing resource wastage compared to traditional protection mechanisms.

Further improvements in dedicated traffic-aware strategies were proposed in [63], integrating DPP with traffic forecasting models to reduce bandwidth blocking probability, especially under dynamic traffic demands. Additionally, [64] investigated the integration of DPP with Content-Oriented Networking (CON) for content delivery in EON, evaluating its impact on capital expenditure (CAPEX), power consumption, and spectrum usage. Numerical analyses based on Cisco traffic patterns and real-world network topologies indicate that DPP protection introduces an additional overhead of 2.1 to 3.5 times in the examined metrics. However, optimizing the content delivery scenario can lead to up to 50% savings across key network performance metrics. Overall, integrating DPP with CON presents a promising approach for efficient optical bandwidth distribution in next-generation networks.

Next is SBPP where, multiple protection paths share common resources, reducing the overall spectrum and cost requirements for survivability. As illustrated in Fig. 3, the solid lines represent primary paths, while the dashed lines indicate the shared backup paths. In this example, the primary paths S-A-B-D and S-E-F-D each have separate working paths but share a common backup path, S-C-D. If a failure occurs along the primary path S-A-B-D, the backup path S-C-D is activated. However, SBPP follows a key constraint: only one failure can be handled at a time per shared backup resource. If a failure occurs

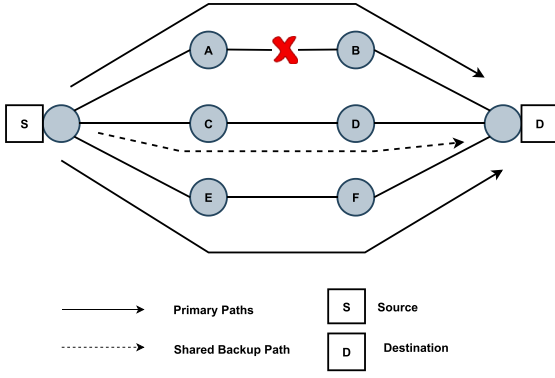


Fig. 3. Shared Backup Path Protection scheme.

within the shared backup path itself, all primary paths relying on that backup must be rerouted, requiring the establishment of new shared backup paths.

While SBPP is more complex to provision and maintain, it allows multiple primary paths (N different pathways) to share the same protective path, significantly reducing the spectrum and cost requirements. However, because backup resources are shared, SBPP does not guarantee full traffic recovery in all failure scenarios, particularly if simultaneous failures occur. In contrast to DPP, where each backup lightpath has exclusive spectrum resources, SBPP allows backup lightpaths from different connections to share resources, provided that only one failure occurs at a time. This key difference makes SBPP more spectrum-efficient and cost-effective than DPP [65].

SBPP has been widely explored in the literature. The authors in [66, 67] developed SBPP using Integer Linear Programming (ILP) to minimize the number of wavelength links and wavelength converters, optimizing network cost. A heuristic routing approach for shared protection provisioning was proposed in [68], where simulations compared network utilization between dedicated and shared protection techniques. Further, [69] introduced novel heuristic algorithms, such as Adaptive Frequency Assignment with SBPP (AFA/SBPP) and Most Sub-Carriers and Average Longest Path First (MSALPF), alongside existing RSA algorithms adapted for SBPP scenarios. Their result in Fig. 5 demonstrated that SBPP enables a 28% reduction in spectrum usage compared to DPP, highlighting its efficiency in optical networks.

Other studies have examined the trade-offs between SBPP and 1+1 protection. The authors in [70] developed models to minimize spare capacity requirements and optimize link frequency slot (FS) utilization by considering transponder tunability. In [71], efforts were made to reduce spectrum consumption by proposing a Routing, Modulation, and Spectrum Assignment (RMSA) algorithm, called Lowest Cost First and Shortest Path Algorithm (LCF), to optimize resource allocation for both working and backup lightpaths.

Additional research has focused on enhancing SBPP performance through spectrum management techniques. A study in [72] introduced Shared Path Protection with Reconstructing Shareable Bandwidth based on Spectrum Segmentation (SPP-RSB-SS). This approach segments the spectrum to reduce fragmentation and increase spectrum allocation success rates. Another study in [73] explored SBPP models that provide protection against link or node failures in survivable virtual network embedding over elastic optical networks.

For advanced survivability optimization, researchers in [74] proposed an Integer Linear Programming (ILP) model for Survivable Routing, Modulation, and Spectrum Allocation (SRMSA) in Adaptive EON, referred to as Blocking Aware Survivable Path Provisioning (BASPP). This model enhances survivability by sharing backup resources when spectrum sharing is not possible, utilizing methods like Backup Spectrum Reservation with Multi-Path Protection (BSR-MPP) and SBPP. Additionally, in [75], Backup Spectrum Reservation with Multiple Backup

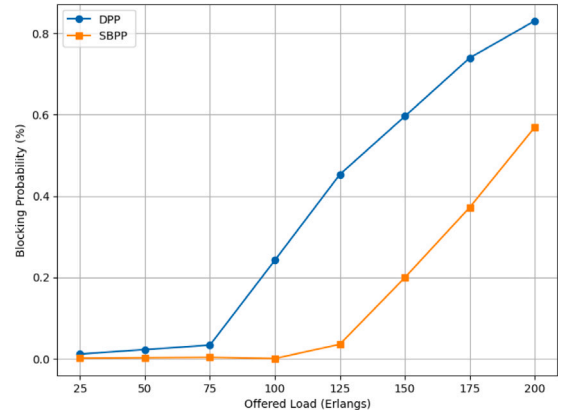


Fig. 4. Comparison of SBPP and DPP scheme using USNET topology.

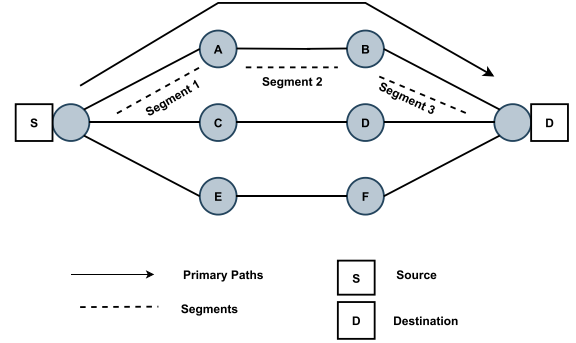


Fig. 5. Segmented protection and restoration scheme.

Path Protection (BSR-MPP) was introduced to increase backup spectrum sharing by implementing reserved backup spectrum and multi-path protection strategies.

Overall, SBPP presents a highly efficient approach to survivability in EON, balancing resource optimization with protection capacity. While it reduces spectrum and cost requirements compared to DPP, it introduces greater complexity in provisioning and cannot always guarantee full traffic recovery in all failure scenarios. However, with the integration of advanced heuristic algorithms and optimization models, SBPP continues to be a promising strategy for future optical network resilience.

To evaluate the efficiency of SBPP and DPP, we conducted simulations using blocking probability as the key metric for assessing network efficiency, as it reflects the ability to accommodate new connections under limited spectral resources. To evaluate DPP and SBPP, we conducted simulations using the USNET topology, a realistic model for long-haul optical networks. Our event-driven simulation employs a Poisson arrival process and exponential holding times to dynamically model connection establishment and release. Results show that SBPP reduces blocking probability by approximately 30% compared to DPP, demonstrating its superior efficiency in resource allocation and network performance as shown in Fig. 4.

2.2. Segmented protection and restoration schemes

A survivability strategy called Segmented protection and restoration scheme as shown in Fig. 5 is used in elastic optical networks to increase network resilience and guarantee continuous operation in the event of failures. With this plan, the network is divided into smaller segments, each of which has a subset of nodes and the links that connect them. The objective is to minimize the overall disturbance to network traffic by localizing the impact of failures by limiting them inside smaller

segments for better understanding. Using this as an example, a series of connected links along the main lightpath is referred to as a primary segment. In other words, a segmented protection lightpath can only be deployed when a component along the primary segment experiences a failure and only covers the primary segment. The advantage of a segmented protective lightpath is shown in Fig. 5. It is necessary to establish an N -connection from the source (S) to the destination (D). The three links that make up the main lightpath are links S-A, A-B, and B-D. Here, a principal segment is made up of the links S-A, A-B, B-D and their end nodes. Segmented routing was first described in publications [76,77], where the authors established SR technology in IP networks where multi protocol label switching (MPLS) is utilized. Subsequently, this SR approach is employed in the optical network fields to preserve and increase survivability. Here, MPLS is nothing but a protocol that was created to swiftly and effectively transport data packets to their intended locations. It is preferable than standard Internet Protocol (IP) routing because it transmits data directly to its destination rather than bouncing it around the internet before sending it to its intended location. With the help of the OpenFlow protocol [78–82] and GMPLS operations [42,45,83,84], SDN mostly employs the SR approach. With the help of SDN, we will outline the developments we can make or that are currently occurring in the field of survivability in the future research part [85]. By utilizing traffic grooming approach, author in paper [86] developed segment protection algorithm for working path to reduce service blocking ratio as much as possible.

2.3. Path and span restoration schemes

In SEON, PR [70] and SR [87] are two distinct techniques used to recover disrupted traffic and maintain network resilience in the event of failures. PR restores the entire communication path or lightpath affected by a failure, ensuring end-to-end service continuity. Since a communication path consists of a sequence of interconnected nodes and links, PR reconstructs the complete route from source to destination.

In contrast, SR focuses on restoring only the failed segment rather than the entire path. A span consists of a single connection or multiple consecutive links between two adjacent nodes. Instead of rerouting the entire traffic flow, SR repairs only the affected segment, making it a faster and more resource-efficient solution for localized failures. Various studies have explored SR techniques to optimize resource allocation. The authors in [88] investigate different spectrum conversion capabilities to minimize spare capacity requirements and the maximum number of link frequency slots. Additionally, techniques such as Bandwidth Squeezed Restoration (BSR) [61,89,90] enhance restoration efficiency by adapting to the limited frequency slot capacity on each fiber link. Another study [91] applies SR to CO-OFDM-based EON, leveraging spectrum conversion to further reduce spare capacity needs and maximize traffic restoration. A novel fiber-optic modulation approach proposed in [92] achieves high spectral efficiency, flexible bandwidth allocation, and resistance to impairments like chromatic dispersion, aligning with the findings in [91]. Moreover, spectrum conversion is highlighted as a key factor in reducing spare capacity redundancy and enhancing spectrum efficiency in flexi-grid EON.

The choice between PR and SR depends on failure type, network topology, traffic patterns, and required restoration speed. PR is more suitable for large-scale failures that impact the entire communication path, ensuring complete service recovery. Conversely, SR is faster and more effective for localized failures, as it repairs only the damaged span without affecting the entire path. In practice, many networks employ a combination of both techniques to provide comprehensive protection and restoration coverage, balancing speed, efficiency, and resource utilization.

To further analyze the effectiveness of PR and SR, we conducted a dynamic simulation of an EON using the USNET topology. Both models were implemented within an event-driven framework, where connection requests followed a Poisson arrival process with exponential

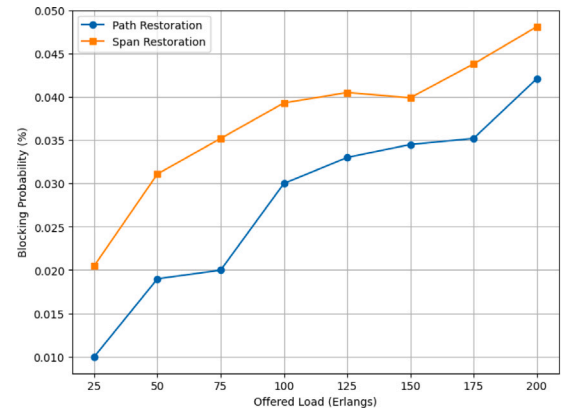


Fig. 6. Comparison of Path and Span restoration scheme using USNET topology.

holding times. Each connection required 100 Gb/s (equivalent to 4 contiguous slots) and was initially routed along a primary path. In the PR scheme, a failure triggered a complete end-to-end re-routing attempt for the affected connection. Meanwhile, the SR scheme aimed to optimize local recovery by re-routing only the failed span, using a k-shortest paths algorithm to find alternative routes. Since not all connections experience failures, the restoration process was applied selectively to a fraction of active connections. While SR was expected to reduce blocking probability, our simulation results showed higher blocking in span restoration compared to path restoration. This was primarily due to higher congestion in alternative local routes, which offered limited path diversity. Additionally, our simulation did not explicitly account for spectrum fragmentation, a well-known challenge in elastic optical networks, further influencing blocking probability. When averaging results across moderate and high traffic loads, the PR scheme achieved a 25% lower blocking probability compared to SR, demonstrating its superior efficiency in maintaining network resilience as shown in Fig. 6.

2.4. Ring cover and p-Cycle protection schemes

To enhance network survivability and ensure efficient failure recovery, SEON employ two advanced protection techniques: Ring Cover Protection and p-Cycle Protection [93]. These methods are designed to mitigate failure impacts while maintaining uninterrupted service delivery. Ring Cover Protection involves deploying specialized backup rings alongside primary communication paths, forming circular protection routes that instantly take over in case of failure. In contrast, p-Cycle Protection is a more flexible and resource-efficient approach that pre-computes backup paths (p-cycles), allowing for rapid adaptability to different failure scenarios.

Ring Cover Protection can be further classified based on network topology into UPSR and BLSR, both of which optimize spare capacity sharing [8]. The concept was first introduced in EON in [95] to reduce overall spectral usage and minimize required protection capacity. A case study in [95] demonstrates the implementation of Ring Cover Protection in an EON topology, where two pre-deployed rings, (B-F-E-C-B) and (B-A-F-E-C-B), serve as backup paths in case of failures. If a failure occurs on the B-C link, it disrupts operational connections between node pairs B-C and B-D. To restore these flows, the ring (B-F-E-C-B) allocates spectrum slots 3–4 for the B-C flow, while the ring (B-A-F-E-C-B) assigns spectrum slots 5–7 for the B-D flow. Ensuring spectrum continuity is crucial for effective ring based recovery, requiring two pairs of opposite fibers on each link. As illustrated in Fig. 8, dual opposite fiber pairs (e.g., link D-E) enable the reuse of the same spectral resources on different fibers, allowing both working flows and

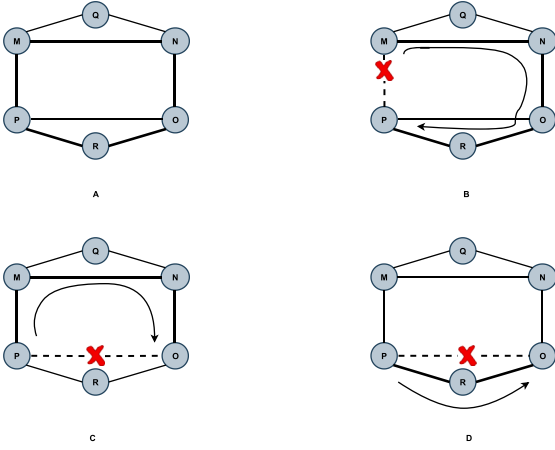


Fig. 7. (A) One illustrative instance of a p-cycle. (B) Protecting on-cycle links. (C), (D) Protecting straddling links [94].

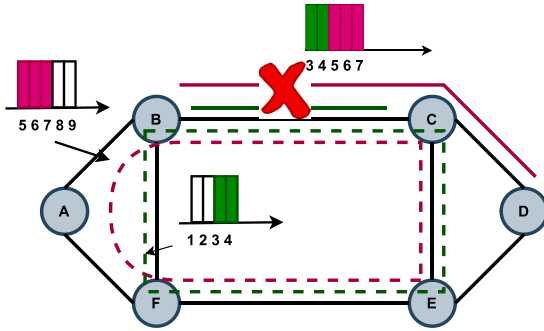


Fig. 8. One illustrative instance of ring coverage within an elastic optical network [95].

protection rings to function seamlessly. This method significantly enhances network resilience while efficiently utilizing spectral resources in SEON.

On the other hand, p-Cycle Protection technique has been widely studied as a robust mechanism for enhancing network survivability in SEON. The pioneering work in [87] introduced p-cycles as a means to improve restorable network design and resource utilization, significantly advancing the understanding of pre-configured protection structures.

To illustrate, consider the example shown in Fig. 7, where multiple p-cycles of varying lengths (from one to five links) can be identified, including M-Q-N-M, M-Q-N-O-P-M, M-N-O-P-M, M-N-O-R-P-M, and P-O-R-P. While some cycles (e.g., M-Q-N-M and P-O-R-P) cover all nodes, they are deemed unacceptable because they do not encompass all links. In contrast, cycles such as M-Q-N-O-P-M include all links, making them suitable for network protection. However, the M-Q-N-O-R-P-M cycle, despite covering all nodes and links, is unsuitable due to the imposed maximum cycle length constraint of five links [94]. The application of p-cycle protection has been further expanded in [96], where researchers extended the technique to protect transparent optical express flows in regional networks. This study also compared p-cycle protection with SBPP in terms of capacity efficiency. Other studies [97–99] explored the Failure Independent Path-Protecting (FIPP) p-cycle algorithm, leveraging heuristic-based techniques to enhance path protection in elastic optical networks. Further improvements were made in [86], where a multiplexing algorithm was used to optimize backup resource utilization in FIPP p-cycle protection.

Under cloud-ready SEON with SDN control and anycast relocation, stream-based, linear pre-configured protection has been proposed as an

alternative to cyclic schemes. The approach in [100] enables relocation-aware protection by first transforming the physical graph with a virtual hub connected to all data centers; two link-disjoint paths from the source to the hub are computed and then mapped back to one or two data centers, allowing relocation to a nearer center while keeping BV-OXC pre-configured along a non-divergent linear overlap of backup routes for rapid switchover. The same work categorizes FIPP p-cycle operation and explains how a stream structure avoids branch-selection ambiguities when the primary is on or partly on the cycle, while remaining distinct from PXT trails—which can be infeasible in cloud-ready SEON if a trail traverses multiple data centers.

Complementing this, [101] develops fragmentation-aware resource-efficient protection with fast recovery (FRPFR), which merges backup paths into pre-configured streams and selects spectrum gaps using a fragmentation-minimizing contiguous-gap rule under contiguity/continuity constraints; it also applies a relocation-enabling topology transform and provides polynomial-time bounds for controller deployment. Taken together, these stream-based designs preserve immediate failover while aligning with anycast relocation and fragmentation control, offering a flexible alternative to p-cycle families in cloud-ready SEON settings [100,101].

Comparative studies between Ring Cover Protection and p-Cycle Protection have also been conducted [102,103], evaluating their effectiveness in EON survivability. To further reduce spectrum utilization and spectrum fragmentation ratio (SFR), researchers in [104] introduced Spectrum Shared pre-configured cycles (SS-p-cycles), allowing multiple p-cycles to share spectral resources along common links. Additionally, [105] proposed a p-cycle-based protection strategy for multicast traffic demands, further broadening its applicability. Recent advancements in genetic algorithms have also influenced EON survivability techniques. In [106], researchers introduced the GPCPS, employing an EGA to address issues such as spectrum fragmentation and redundancy in p-cycle protection systems. In [107], three distinct directed p-cycle designs were proposed to minimize power consumption and spectrum utilization in EON. Additionally, [108] introduced two novel link-based p-cycle evaluation methods: individual p-cycle selection and p-cycle set selection, to identify optimal p-cycle configurations for failure-dependent link protection. Other studies have explored energy efficiency and power optimization in p-cycle protection. In [109,110], the authors established p-cycles as a technique to reduce energy consumption in EON. Meanwhile, in [111], a p-cycle reconfiguration strategy was proposed to dynamically optimize protection structures, improving spectrum efficiency and reducing bandwidth blocking probabilities. Overall, p-cycle protection continues to be a promising failure recovery mechanism in EON, offering high-speed restoration, resource optimization, and adaptability to different network scenarios.

To further assess the effectiveness of Ring Cover Protection and p-Cycle Protection, we conducted a dynamic simulation using the USNET topology, a realistic model for wide-area optical networks. Connection arrivals followed a Poisson process, with exponential holding times spanning 1000 time units. Each connection, requiring 4 contiguous spectrum slots to support 100 Gb/s under BPSK modulation, was assigned a primary path using a shortest-path algorithm, while backup paths were pre-configured through ring cover or p-cycle protection schemes. In our evaluation, the Ring Cover Protection scheme selects the longer segment of the ring, making it less resource-efficient. In contrast, the p-Cycle Protection scheme selects the shorter segment, optimizing backup resource consumption. This technical approach rigorously enforces continuity and contiguity constraints across links, ensuring accurate resource allocation. Our results confirm that p-Cycle Protection is significantly more efficient than Ring Cover Protection, achieving an overall blocking probability reduction of 42% shown in Fig. 9. This demonstrates superior network resilience and resource optimization in SEON.

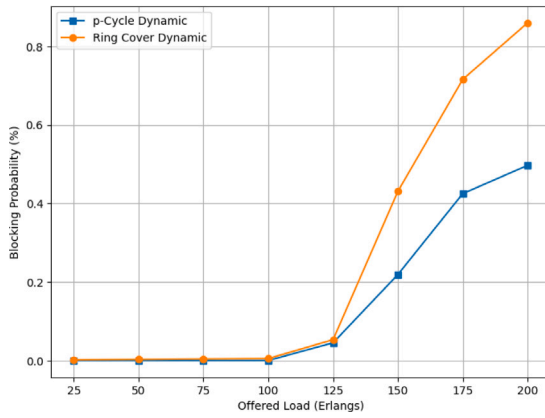


Fig. 9. Comparison of p-cycle and Ring protection scheme using USNET topology.

3. Different failures scenarios

Failures in EON significantly threaten service continuity and demand robust survivability mechanisms. These failures manifest in various forms. Link failures disrupt communication routes and require rapid rerouting; node failures cause connectivity loss and potential network congestion; transponder malfunctions degrade signal quality; and wavelength conflicts introduce signal interference that reduces reliability. More severe disruptions such as fiber cuts lead to total transmission loss, while simultaneous failures introduce high complexity, requiring advanced restoration strategies. Additional risks include power outages, optical signal degradation, hardware or software faults, routing errors, and security attacks, all of which can compromise network performance or availability.

To understand the scope and relationship between these failures, Fig. 10 presents a failure classification tree, categorizing them into single link failures, multi-link failures, and path failures. Single link failures often result from localized issues like hardware faults or isolated node failures. Multi-link failures represent more extensive disruptions, commonly caused by natural disasters or simultaneous equipment failures. Path failures occur when an entire communication path becomes unusable due to cascading link or node disruptions. This hierarchical classification is essential for designing effective and scenario-specific protection and restoration mechanisms within the EON architecture.

Informed by an extensive literature review and domain knowledge, this section provides a detailed analysis of these failures to support the development of resilient mitigation strategies. Key failure scenarios include:

3.1. Single and dual-link/multi-link failure scenarios

Link failures in SEON pose significant challenges to network resilience and uninterrupted service delivery. A link failure occurs when a physical fiber link between two nodes becomes non-functional, which can result from fiber cuts, device malfunctions, or signal degradation. Such failures disrupt connectivity, reducing the network's ability to maintain operational efficiency. Several studies [55,62,65,69,93,112–116] have explored single-link failures, discussing protection and restoration strategies to enhance EON survivability.

In contrast, multi-link (or dual-link) failures are significantly more complex. These failures involve the simultaneous failure of multiple fiber links, affecting multiple communication pathways across the network. Unlike single-link failures, which impact a specific path and trigger localized rerouting, multi-link failures introduce greater interdependencies among lost links, requiring sophisticated restoration strategies. The key challenge lies in managing the increased complexity, limited

available resources, and network-wide disruptions. Researchers [117–128] have extensively analyzed multi-link failure scenarios, proposing various solutions to mitigate their impact.

To address multi-link failures, advanced protection and resource allocation mechanisms are essential. One approach, introduced in [126], leverages Software-Defined Networking (SDN) to enhance network survivability. The study proposes an Adaptive Survivability (AS) strategy, which optimizes the balance between path protection efficiency and routing costs. Additionally, [129] introduces a novel Routing, Modulation, and Spectrum Assignment (RMSA) algorithm, optimizing both anycast and unicast traffic flows. The performance of RMSA algorithms and the effectiveness of the AS approach are further evaluated in [130], demonstrating their potential in improving SEON resilience under multi-link failure conditions.

3.2. Node failure

In SEON [8], a node failure occurs when a network node—responsible for routing, forwarding, and processing data—malfunctions or becomes non-functional. Such failures disrupt connectivity, impacting all communication paths passing through the affected node. This can lead to service interruptions, degraded network performance, and a poor user experience.

To mitigate the impact of node failures, SEON employ advanced restoration techniques, including traffic rerouting via alternative paths that bypass the failed node. Additionally, backup resources, distributed control mechanisms, and intelligent restoration procedures ensure continuous network operation even in failure scenarios. These mechanisms allow SEON to adapt and recover promptly, maintaining high network availability and reliable service delivery. Unlike single or multi-link failures, node failures are significantly more complex and costly since a single node failure can lead to multiple link or path failures. Due to this complexity, research often focuses more on link failure scenarios, as seen in the previous discussions on single and multi-link failures. However, studies in [111,112,128,131,132] specifically address node failure scenarios in SEON, analyzing their challenges and proposing solutions to enhance network survivability and resilience.

Beyond link and node failures, another critical failure scenario in SEON is path failure, where an entire communication path becomes non-operational. The next section will explore path failures, their causes, and effective strategies for ensuring survivability in SEON.

3.3. Path failures

In SEON [8], a path failure occurs when a critical communication route comprising interconnected links and nodes are disrupted due to various causes such as fiber cuts, hardware failures, or unforeseen events. This disruption severs connectivity between the source and destination nodes, affecting services and data flows that rely on the impacted path.

To mitigate path failures, SEON utilize resilient protection and restoration mechanisms, including pre configured backup paths and dynamic rerouting strategies. By promptly detecting failures and adapting to changing network conditions, SEON minimize service disruptions and ensure high network availability. This proactive approach enhances network resilience, reinforcing the system's ability to deliver reliable and continuous connectivity to users. Path failures often arise from link failures [55,62,65,69,93,112–116], multi-link failures [117–128], or node failures [111,128,131,132]. Since a path consists of multiple links and nodes, any failure within these components can directly or indirectly impact the entire path. Researchers have also explored additional factors contributing to path failures in studies [65,70,111,127,128,131–139] emphasizing the need for robust path protection and restoration strategies to safeguard network performance.

Beyond path failures, another critical challenge in SEON is hardware failure, which can severely impact network operations. The next section will discuss hardware failures, their implications, and strategies for ensuring network survivability in SEON.

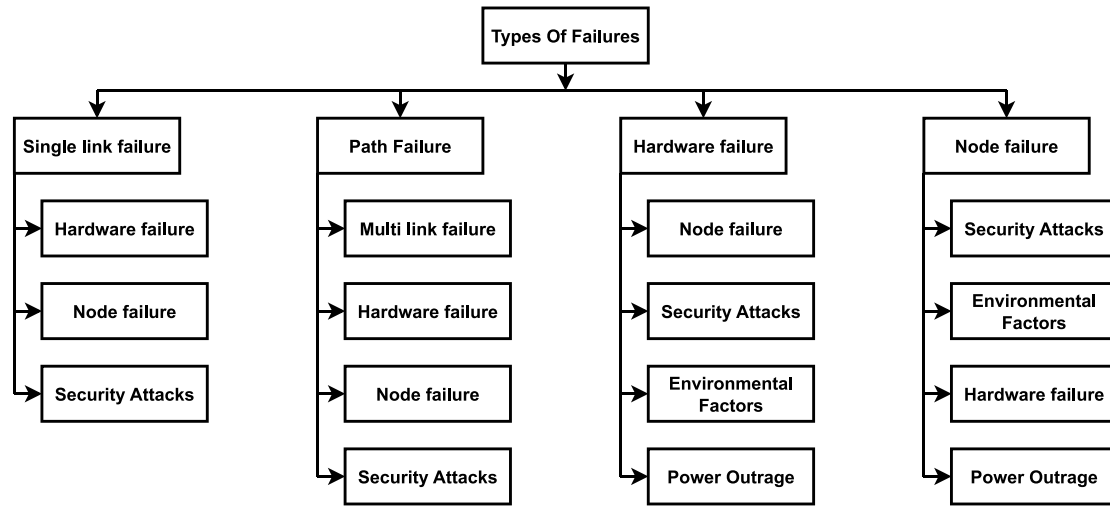


Fig. 10. Types of failure covered by our survey.

3.4. Hardware failures

Hardware failures pose a significant challenge in SEON, as they can disrupt network operations and compromise service availability. These failures affect various critical network components, including optical switches, transponders, routers, amplifiers, and other key infrastructure elements. Hardware malfunctions can result from factors such as manufacturing defects, wear and tear, overheating, electrical failures, or other operational issues.

Single/multi-link and node failures are also categorized as hardware failures, as they often stem from physical component malfunctions. In this section, we explore various optical devices and review research efforts addressing hardware failures in the context of SEON [8]. Some of the key hardware components used in EON include:

- **ROADM [140–143]:** It is an essential component of flexible optical networks. Without precise knowledge of traffic growth trends, ROADM enables network operators to dynamically route and swap wavelength channels through software control resources. Specific optical components, such as WSS [144], Spectrum Selective Switches (SSSs), multicast switches (MCSs), and various transponder combinations, are used to create ROADM designs. However, the flexibility and performance needs are what determine which of these components to use and in what combinations. Furthermore, the flexibility, functionality and reconfigurability of optical node operations are further increased by ROADM systems with colorless, directionless, and contentionless (CDC) characteristics [145–153].
- **WSS [144]:** A WSS consists of a switching array that manipulates light signals that have been dispersed in terms of wavelength, without the need for physically separating the dispersed light into distinct ports. The setup described is sometimes referred to as a “disperse and switch” arrangement. There exist primarily two kind of WSSs; 1X2 WSS and 1XN WSS [154].
- **SBVT [53]:** The main component of EON, which contributes to its adaptability and effectiveness, is the SBVT shown in Fig. 14. The SBVT consists of a group of virtual BVT shown in Fig. 14 [155,156] with reduced capacity. The multi-flow transponder is capable of accommodating numerous optical flows with varying data rates, spanning from 10 Gb/s to 1 Tb/s. These flows are directed simultaneously in multiple directions. The system provides support for adaptive modulation formats that can dynamically modify the distance at which they are applied. A collection of sub-channels is produced at the output of the SBVT. These sub-channels are then transmitted over media channels in order to be directed toward various directions [157–161].

- **EDFA [53]:** The EDFA is a device utilized for the amplification of optical fiber signals. This technology finds application in the telecommunications industry as well as in many research domains. The EDFA incorporates erbium as a dopant material. The term “doping” pertains to the use of chemical elements in order to enhance outcomes by manipulating electrons [155,162–174].

Efforts to mitigate hardware failures in SEON are continuously evolving, with advancements in hardware reliability, real-time monitoring, and network management strategies. These improvements contribute to the overall resilience and survivability of EON in the face of hardware-related challenges.

To assess the impact of failures on network performance, we conducted a fully dynamic EON simulation that enforces key network constraints. Each link consists of 320 contiguous slots, with separate arrays for primary and backup allocations, ensuring slot continuity and contiguity. Connection arrivals follow a Poisson process, with exponential holding times, while four failure scenarios—single link, multi-link, hardware, and node failures—are simulated. Recovery events are scheduled after a fixed recovery time, and fragmentation effects are not modeled to isolate the direct impact of failures.

In the hardware failure scenario, we simulate device degradation, where optical devices (e.g., transponders, optical amplifiers) deteriorate due to aging, thermal stress, or environmental conditions, leading to reduced spectral capacity. For example, an optical amplifier exposed to high temperature fluctuations may suffer from power loss and increased noise, effectively rendering a fraction of its slots unavailable (modeled as setting them to -1). This behavior, previously analyzed in [175], significantly impacts network performance. Our results show that blocking probability increases with traffic load, rising from 5.7% at 125 Erlangs to 55.0% at 200 Erlangs, emphasizing the severe consequences of hardware degradation.

Other failure scenarios exhibit similar trends. Single link failures, where a random link is removed and later restored, cause blocking probabilities to rise from 0% at low loads to 40.7% at 200 Erlangs. Multi-link failures, which involve simultaneous failures of multiple links, result in blocking reaching 50.8% at 200 Erlangs. Node failures, where an entire node is removed and subsequently recovered, yield blocking probabilities from 0.1% at low loads to 56.2% at 200 Erlangs. These results, as shown in Fig. 11, illustrate that while failure scenarios have minimal impact under low traffic, they cause severe network performance degradation under higher loads.

The goal of our study is to highlight the raw impact of different failures in EON without protection or restoration mechanisms. The observed high blocking probabilities underscore the critical need for

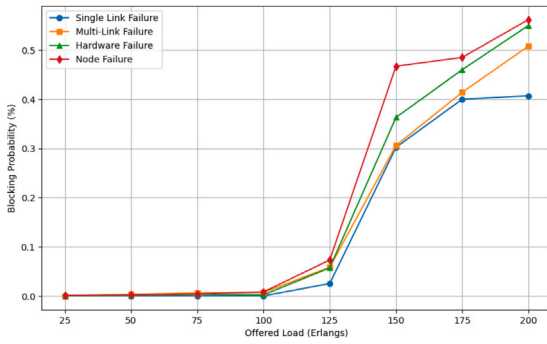


Fig. 11. Comparison of single-link, multi-link, Hardware and Node failures using USNET topology.

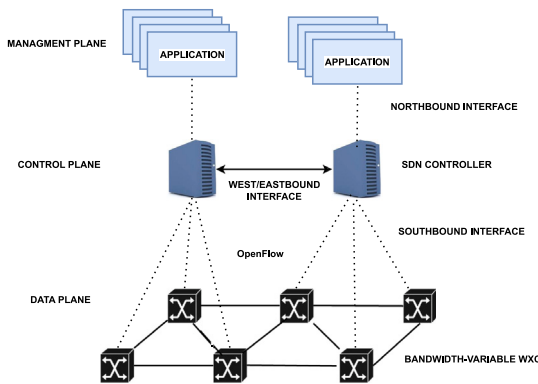


Fig. 12. SD-EON architecture.

robust protection and restoration schemes in real networks, particularly for hardware and node failures, which present the most challenging recovery scenarios (see Table 2).

4. Open challenges and future researches

Based on the findings of our literature review, it is evident that most previous research on SEON has primarily focused on introducing novel concepts and proposing feasible yet straightforward solutions. While these studies have significantly contributed to the field, there remains a need for deeper exploration and further advancements in several key areas. To facilitate future research and expand upon the existing knowledge base, this section is structured into the following critical subtopics: (1) Centralized network control and management (2) Spectrum management considering survivability. (3) AI in SEON. (4) physical layer Impairments. (5) Hardware development. (6) Disaster management. These research directions provide a comprehensive road map for advancing SEON survivability, guiding future studies toward more resilient, adaptive, and efficient optical networks.

4.1. Centralized network control and management

Over the past few decades, Generalized Multi-Protocol Label Switching (GMPLS) [45,83,84,180,181] has become the dominant architectural framework for backbone networks in the telecommunications industry. Its widespread adoption is attributed to flexibility, automatic manageability, fault tolerance, and rapid resiliency, with restoration times of less than 50 ms [182]. GMPLS has successfully replaced SDH/SONET and ATM, establishing itself as the global standard for Layer 2/3 protocols. Offering long-term compatibility with legacy systems, GMPLS supports label-based resource representation, enabling seamless integration with technologies like Frame Relay and ATM.

However, Software-Defined Networking (SDN) is emerging as a potential successor to GMPLS. While GMPLS has yet to achieve full interoperability between IP networks and optical transport networks (OTN) [183], SDN provides greater flexibility and dynamic network management. With advancements such as Bandwidth-Variable Transponders (BVTs) [184,185], EON can allocate frequency slots dynamically based on bandwidth demand, overcoming GMPLS's rigid ITU frequency slot allocation [146].

One of the key challenges in future-proof network architectures is the demand for programmable networks, driven by evolving telecommunications technologies. SDN addresses this need by fully separating the control and data planes, enabling centralized network management that can be physically and logically distributed for unprecedented scalability and flexibility. This has led to the widespread integration of SDN functionalities into modern routers.

The increasing demand for programmable networks in EON has positioned Software-Defined Networking (SDN) as a key enabler of next-generation network architectures. SDN offers centralized control, enhanced flexibility, and dynamic resource management, making it a crucial technology for Software-Defined EON (SD-SEON). To better understand the evolution of SD-SEON, we examine an expanded SDN architecture that integrates control, data, and management planes. The study in [186] proposes a comprehensive SDN-based framework, illustrated in Fig. 12, where the Southbound Interface enables communication between the control and data planes, while the Northbound Interface, based on OpenFlow [15,123,187–189], facilitates interactions between the control and management planes—although standardization remains a challenge. Additionally, the East/Westbound Interface ensures efficient communication between multiple network controllers, enhancing network coordination and scalability [190]. These architectural advancements underscore SDN's potential to transform SEON, but challenges related to interoperability, security, and resource optimization must be addressed for widespread adoption.

In regard to the SD-SEON, we have identified some difficulties that necessitate resolution :

- The identification, detection, and localization of failures in SD-SEON can be significantly enhanced through SDN's centralized control and management capabilities. Unlike traditional networks, where failure detection is often distributed and reactive, SDN enables real-time monitoring and proactive fault management by maintaining global visibility over all links and nodes. By leveraging this centralized intelligence, SDN can efficiently analyze network status, detect anomalies, and precisely locate failures, allowing for faster recovery and improved network resilience [191].
- For centralized control in SD-SEON, a standardized protocol like OpenFlow is essential for managing and controlling the optical data plane. However, current protocols face limitations in handling the unique requirements of optical networks, such as dynamic spectrum allocation, flexible grid management, and ultra-low latency switching. To fully leverage SDN's capabilities in SEON, there is a need for enhanced protocols that support advanced optical functionalities, improved scalability, and seamless interoperability with existing network infrastructures. Future developments should focus on extending OpenFlow or designing new optical control protocols that can efficiently manage elastic spectrum resources, wavelength routing, and fault recovery in SEON.
- SD-EON can rapidly and consistently detect environmental factors that may lead to fiber cuts or physical damage, minimizing data loss and ensuring network resilience. By leveraging centralized monitoring and real-time analytics, SD-EON can identify signal degradations, power fluctuations, or abnormal transmission losses, allowing for immediate failure localization and proactive restoration measures. This capability significantly enhances network survivability, ensuring minimal service disruption in the event of physical infrastructure damage.

Table 2
Comparison table.

Survey, Published year	Traditional Optical Network	EON	Protection	Restoration	Link protection	Link restoration	Path protection	Path restoration	Dpp	SBpp	Ring cover	p-cycle protection	Span protection	Segmented Rerouting	Single link failure	Multi link failure	Node failure	Path failure	Hardware failure	Fragmentation	Focus
[176], 1999	✓		✓		✓		✓		✓	✓					✓			✓			This study introduces and evaluates path and link restoration schemes specifically designed for static traffic demands.
[177], 1999	✓			✓	✓	✓		✓	✓	✓					✓			✓			This study introduces and evaluates path and link protection schemes specifically designed for static traffic demands
[178], 2000	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓				✓		✓	✓	✓		This paper examines the methodologies employed to attain survival in conventional optical networks and explores the advancements in these strategies as they pertain to the enhancement of survivability in next-generation WDM networks.
[179], 2002	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			✓		✓	✓			This paper examines the primary optical protection strategies employed in WDM networks, with a specific emphasis on ring and mesh network configurations.
[63], 2015		✓	✓	✓	✓	✓	✓	✓	✓	✓					✓	✓	✓	✓	✓		This paper aims to provide an overview of the emerging network services that are built upon cloud computing and content oriented paradigms. Additionally, it will discuss the EON designs and the strategies they employ to ensure survivability.
[8], 2016		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓			✓	This study provides an overview of the present state of the art in survivable EON. It conducts a comprehensive assessment of the existing literature to summarize several topics related to the survivability of EON.
This work		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	This work is centered around various survivability schemes and different failure scenarios that have a direct or indirect impact on the survivability of the EON.

- With SDN technology providing complete visibility and control over every node in the optical data plane, SD-EON can implement advanced security measures to protect against cyberattacks. One promising approach is the integration of blockchain technology, which offers decentralized security, data integrity, and tamper-resistant transaction logging. By leveraging blockchain-based authentication and encryption, SD-EON can enhance trust, prevent

unauthorized access, and secure control communications, ensuring robust protection against cyber threats while maintaining network reliability and resilience.

For SD-SEON to remain viable, several critical challenges identified through literature review must be addressed. Given that SDN relies on centralized control and management, ensuring efficient scalability,

security, and fault tolerance is essential. Additionally, advancements in hardware development are necessary to support common protocols like OpenFlow, enabling seamless integration, dynamic resource allocation, and intelligent failure recovery. Overcoming these challenges will enhance network resilience, adaptability, and overall performance, ensuring SD-SEON can meet the evolving demands of future optical networks.

4.2. Spectrum management

Fragmentation in a SEON refers to the partitioning of the available optical spectrum into discrete and non-contiguous segments. This occurs due to the allocation of wavelength channels with varying sizes to accommodate diverse traffic demands, similar to the fragmentation challenges found in EON. However, in a survivable network, fragmentation introduces additional complexities that affect network resilience, fault recovery, and restoration strategies.

This section provides a detailed overview of fragmentation in SEON, focusing on its causes and implications for network survivability. To ensure efficient protection and restoration mechanisms, it is crucial to address the key sources of fragmentation that impact SEON performance. The following are major contributors to fragmentation in SEON that require careful consideration.

- Service Edge Overlay Networks employ protection and restoration schemes to enhance network resilience against link or node failures. However, these mechanisms often lead to fragmentation, as backup path and segment allocations may not align perfectly with the original traffic demands. This mismatch creates dispersed and underutilized spectrum segments, reducing overall spectral efficiency.

Several researchers have investigated the issue of fragmentation in survivable SEON, proposing diverse strategies to mitigate its adverse effects. Notably, studies such as [56–60,145,192,193] and [24] have introduced techniques aimed at optimizing backup resource allocation, thereby improving spectrum utilization while preserving strong protection and restoration capabilities. Addressing fragmentation is crucial not only for enhancing spectral efficiency, but also for reducing blocking probability and ensuring better overall network performance.

Additionally, the work in [130] introduces a novel approach that incorporates modulation-level awareness into traffic splitting, allowing the network to adapt routing and spectrum assignment (RSA) based on real-time signal quality and modulation needs. This technique significantly improves resource utilization and meets QoS constraints. However, it introduces additional control plane complexity due to its reliance on real-time modulation feedback, which may pose scalability challenges in large, dynamic environments. Note that in SEON, QoT and QoS are both essential but serve different purposes. QoT relates to physical-layer performance, ensuring that lightpaths meet acceptable signal quality thresholds such as OSNR and BER. In contrast, QoS refers to network-layer performance, focusing on user-experienced metrics like bandwidth, latency, and reliability. While QoT determines whether a connection can be physically established, QoS ensures that it meets service expectations. Both must be considered together when designing effective survivability strategies in SEONs.

- The variability of traffic demands in a survivable network can be significant, often leading to fragmentation [194]. This occurs when backup resources are allocated for worst-case failure scenarios but remain underutilized during normal network operation. Mitigating fragmentation in survivable networks is a major challenge for network operators and designers, requiring a balance between ensuring network robustness and optimizing resource utilization.

To effectively address spectrum fragmentation in SEON, a range of advanced strategies can be utilized, including segment-based protection and restoration, dynamic resource allocation, and intelligent backup resource reservation. These approaches help localize protection efforts to critical network areas, minimizing inefficiencies during normal operations while enhancing responsiveness during failures. A recent development in this domain is the application of quantum computing to spectrum management. In particular, the work presented in [195] explores a quantum computing based solution for RSA in flexi-grid optical networks. This method formulates RSA as a quantum optimization problem, significantly reducing path computation time and enhancing scalability for future high-speed network demands. While the benefits are promising—especially for real-time, large-scale routing—challenges remain, notably the current limitations of quantum hardware and the scalability of quantum circuits in practical network environments.

By incorporating these fragmentation-aware strategies, SEONs can achieve better survivability, spectral efficiency, and blocking probability performance, ensuring optimized functionality even under dynamic traffic conditions.

With the increasing complexity of modern optical networks, AI has emerged as a transformative tool for enhancing survivability, traffic prediction, resource optimization, and real-time fault recovery in SEON. The following section explores the integration of AI techniques into SEON architectures and highlights their role in intelligent, adaptive network management.

4.3. Artificial intelligence in SEON

Machine Learning (ML) [196] significantly enhances SEON by improving resilience, efficiency, and performance. ML-driven techniques enable proactive resource allocation, intelligent traffic prediction, and rapid fault recovery, transforming SEON management and survivability. Reinforcement learning optimizes resource utilization, while ML algorithms analyze traffic patterns, enabling predictive maintenance and defect detection.

The application of NN further strengthens SEON by recognizing complex traffic patterns, leading to automated network administration, dynamic reconfiguration, and efficient routing. For instance, ML-powered systems can anticipate high-traffic periods, adjust routing to avoid congestion, and reroute traffic in real-time during failures, ensuring uninterrupted service for 100 Gbps traffic demands. By leveraging predictive analytics, ML enables preventive maintenance, identifying at-risk components and scheduling proactive interventions to minimize network downtime.

Recent research has increasingly focused on leveraging ML to enhance survivability in SEONs, offering adaptive, intelligent approaches for optimizing resource allocation, failure recovery, and spectrum management. In [197], a supervised ML algorithm is proposed to estimate the number of required spectrum slots for traffic demands in virtualized elastic optical networks (VONs). This approach improves protection strategies and allocation efficiency, outperforming traditional Integer Linear Programming (ILP) in computational speed and scalability.

In another study [198], a DRL based heuristic is introduced, utilizing dual agents to dynamically adjust both working and protection schemes. This method enhances S-RMLSA policies by learning optimal configurations under varying network conditions.

Complementing these efforts, [199] presents an Ant Colony Optimization (ACO)-based heuristic specifically designed for SDM-EON. It accounts for crosstalk and fragmentation, improving spectral efficiency and resilience. The method is particularly effective in dynamic environments with variable demands and failures, although its performance depends on careful parameter tuning and convergence control, which may limit fast deployment in real-time systems.

Table 3
Failure management [201].

Work	ML Algorithms	Input data	Task	Experiment Environment
[202]	SVM	Value of the predicted indicators using DES	Failure prediction	Software-Defined Metropolitan Area Network
[203]	Naive Bayes	Recording of Stokes parameters	Failure prediction	7 Gbaud QPSK, 1.4 nm-broad filter
[204]	NN	ESD test values	Failure identification	Transport Software Defined Network
[205]	Decision Tree, SVM	ESD test values	Failure identification	VPI software simulation network
[206]	NN	Features extracted from optical spectrum	Failure identification	VPI software simulation network
[207]	Bayesian Network	Received power and Pre-FEC BER	Failures localization and identification	NWDM, 120 Gbps QPSK
[208]	Clustering and DNN	The power of channels	Anomaly detection	Seven-node testbed with six routing paths
[209]	K-means	Max BER and BER trend	Failure localization	30-node, 50-link Spanish Telefonica optical network
[210]	HNN	Alarm sets	Failure localization	Two COST239 and two NSFNET topologies [211]

Additionally, the study in [200] proposes a joint allocation strategy for computing and optical resources in inter-datacenter SEON. By embedding survivability constraints into the resource model, the approach ensures failover support and latency-sensitive communication. Its main strength lies in its end-to-end optimization, though its practical real-time implementation may be challenged by high computational demands.

Together, these ML-driven strategies significantly enhance S-RMLSA optimization, reduce blocking probability, and improve network resilience. As SEON evolve, the integration of AI and ML continues to play a vital role in developing self-adaptive, scalable, and survivable network infrastructures.

By integrating ML or AI techniques, SEON can become more intelligent, adaptive, and fault-tolerant, providing high-performance, automated, and resilient optical networking solutions (see Table 3).

Ensuring network performance awareness is crucial for effectively managing and controlling optical networks. OPM provides key performance metrics related to signal impairments and degradations, which help in estimating the QoT at the lightpath level for resource assignment and service restoration. Since failures in optical networks can lead to significant data loss and performance degradation, effective failure detection and localization are essential for improving network resilience.

Failure management techniques fall into two broad categories: reactive approaches, where network operators respond to alarms after failures occur often resulting in delays and data loss and proactive approaches, which focus on predicting and preventing failures to allow sufficient time for restoration efforts [212]. ML plays a crucial role in proactive failure detection, as it can identify patterns in network behavior and anticipate failures before they occur [213].

Various ML-based techniques have been proposed for failure prediction and detection. In [202], a SVM based model predicts board failures in Software-Defined Metropolitan Area Networks, using DES to anticipate performance degradation. Similarly, work in [203] employ a Naïve Bayes classifier to detect mechanical stress on optical fibers, categorizing failures such as bending, shaking, or impact. Their technique, implemented in a coherent transceiver, achieves high accuracy with minimal complexity and has been tested in real-time restoration route allocation [214]. In Transport SDN (TSDN), work in [204] propose an Artificial Neural Network (ANN)-based proactive failure detection model, where optical power measurements are preprocessed using the

Extreme Studentized Deviate (ESD) test, reducing computational complexity while improving failure classification accuracy. Additionally, work in [206] utilize Bayesian Networks (BNs) to classify failures due to tight filtering and interchannel interference, allowing dynamic lightpath adjustments to mitigate BER degradation.

ML techniques have also been explored for large-scale failure analysis and localization. The work in [208] proposes a deep neural evolution network (DNEN)-based alarm analysis method for WDM networks, outperforming the SVM and DNN models in fault localization accuracy. work in [209] employ K-means clustering to visualize BER performance trends, assisting operators in diagnosing failures. Additionally, work in [210] introduce a Hopfield Neural Network (HNN)-based method for multi-failure localization in heterogeneous 5G optical-wireless networks [215], achieving high accuracy and rapid response times.

Despite its advantages, ML adoption in optical networks presents reliability challenges, as ML models often operate on a best-effort basis without performance guarantees. Potential solutions include implementing performance monitoring systems with degradation alerts and periodic model evaluation to ensure continuous reliability. Additionally, ML-based automation should be designed to support manual intervention capabilities, ensuring human oversight in critical decision-making. Addressing these challenges will enable ML-driven optical networks to achieve higher efficiency, reduced failure impact, and improved survivability, paving the way for self-healing, intelligent network architectures. The next section explores key physical impairments and their impact on network survivability and performance.

4.4. Physical layer impairments

PLI [216–226] in SEON refer to a range of optical signal degradations that occur during light signal transmission across the network. These impairments include dispersion, interference, noise, and nonlinear effects, all of which accumulate over distance and degrade the QoT [219]. Such impairments can significantly impact the reliability and efficiency of optical communication systems, leading to higher BER, reduced spectral efficiency, and increased signal distortions.

In SEON, effectively managing and mitigating PLI is essential for ensuring the network's resilience, failure recovery, and service availability. Several prevalent PLI types have been identified in the literature, each presenting unique challenges to network survivability. Addressing

these impairments requires further investigation, particularly considering physical layer constraints, to enhance the performance, robustness, and efficiency of EON. The next section explores key PLI challenges and potential strategies for their mitigation.

- In SEON, it is common for working and protective lightpaths to traverse different physical links, often resulting in varying distances and PLIs. These discrepancies can lead to differences in OSNR [227–230] between the primary and backup routes, affecting signal quality and overall network performance. Balancing network survivability and signal quality is crucial in SEON. Route selection, protection scheme design, and PLI mitigation strategies must be carefully optimized to ensure that backup lightpaths maintain the required QoT, minimizing degradation during failure recovery. Network operators and designers must consider these factors to uphold the desired levels of performance, reliability, and resilience in SEON deployments.
- The regeneration of optical signals [8,231–233] in long-haul or high capacity links poses significant challenges due to its complexity and resource requirements. Signal regeneration is crucial for preserving signal integrity and ensuring reliable data transmission over long distances, as optical signals degrade due to attenuation, dispersion, and nonlinear effects. In long-distance and high-capacity optical networks, strategic placement of regeneration sites, efficient resource allocation, and adaptability to dynamic network conditions are essential for maintaining QoT and network reliability. By optimizing regeneration points and integrating flexible signal processing techniques, network operators can enhance SEON survivability, ensuring consistent service quality across extended transmission distances.
- While PLI are primarily managed to ensure signal transmission quality, they can also be exploited for eavesdropping and security breaches in optical communication networks. Unintended signal leakage, interference, and nonlinear effects may inadvertently expose sensitive information, making networks vulnerable to malicious attacks. Managing PLI-related security risks in Survivable EON is a complex and multifaceted challenge. To strengthen optical network security, a comprehensive approach integrating technological safeguards, security policies, and proactive monitoring is essential. Network operators must continuously assess and enhance security protocols to mitigate emerging threats and vulnerabilities, ensuring robust protection against both accidental signal leaks and deliberate attacks.

Effectively addressing PLI in SEON requires a holistic strategy that integrates advanced network planning, continuous monitoring, adaptive management techniques, and seamless coordination across network layers and equipment vendors. As SEON evolve to accommodate diverse and dynamic traffic demands, the efficient management of PLI will remain a critical factor in ensuring network reliability, security, and performance. By implementing proactive mitigation strategies, optimizing resource allocation, and fostering interoperability among network components, operators can enhance SEON survivability and efficiency in the face of growing technological challenges.

4.5. Hardware development

The evolution of hardware components in SEON requires overcoming various technical challenges stemming from the complex and dynamic nature of these networks. Several key difficulties must be addressed to enhance performance, scalability, and resilience. These challenges can either be mitigated or fully resolved through innovations in optical hardware, including enhanced transponders, flexible spectrum allocation technologies, and energy-efficient switching components. As SEON continue to advance, hardware optimization will play a pivotal role in ensuring efficient resource utilization, improved fault tolerance, and seamless adaptability to evolving network demands.

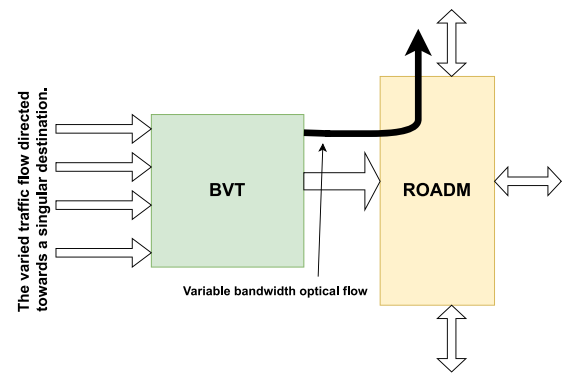


Fig. 13. Bandwidth variable transceiver [234].

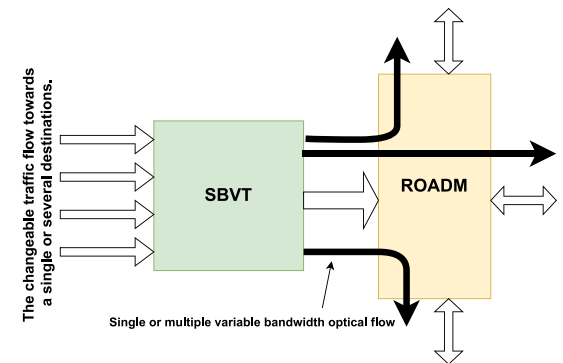


Fig. 14. Sliceable bandwidth variable transceiver [234].

- Power efficiency [235–240] is a critical factor in the design and operation of SEON, especially as network scale and capacity demands continue to grow. Striking a balance between power efficiency, network performance, and reliability presents a complex challenge in SEON development. However, with technological advancements and the integration of energy-efficient hardware, network operators can effectively meet increasing capacity requirements while minimizing energy consumption and operational costs. Optimizing power-aware resource allocation, energy-efficient routing, and intelligent traffic management will be essential for ensuring sustainable and cost-effective SEON deployments.
- The advancement of SEON has facilitated the aggregation of optical signal carriers, commonly known as Network Media Channels (NMCs). This capability paves the way for more flexible and dynamic spectrum allocation, especially under failure conditions. To further strengthen protection and restoration strategies, several innovations and extensions have been introduced to enhance both network resilience and spectrum efficiency. One of the most notable developments in this domain is the application of BVTs [184]. BVTs (see Fig. 13) play a pivotal role in enabling adaptive modulation and bandwidth allocation, making them highly effective in both protection and restoration processes. Their flexibility allows the network to dynamically adjust resources according to the severity of the failure and the traffic requirements, thereby improving survivability without excessive resource consumption. In addition, the work presented in [130] introduces an energy-aware strategy for survivable EON through asymmetrical traffic splitting. This model intelligently distributes traffic between primary and backup paths by considering traffic demands and power consumption profiles. While the approach demonstrates improved energy efficiency and enhanced fault tolerance [100], it may trade off spectral efficiency

under certain network conditions where imbalances in traffic splitting lead to suboptimal slot utilization. Collectively, these advancements highlight the importance of resource-aware survivability mechanisms that balance energy consumption, bandwidth efficiency, and protection robustness—essential elements for the future scalability and sustainability of SEON.

BVTs offer the capability to compress or segment NMC signals through bandwidth squeezing or slicing techniques [55,61]. By incorporating SBVTs [157–160,185,241] SEON can achieve greater spectral efficiency and improved resource utilization. One proposed enhancement is the introduction of a bandwidth reduction flag in the NMC identifier, indicating the use of BVT devices for adaptive bandwidth allocation. This mechanism allows the NMC to dynamically adjust pre-allocated protection resources or restoration capacity [56], further optimizing network performance and survivability.

To overcome the challenges associated with SEON hardware development, fostering collaboration among equipment vendors, network operators, and standardization organizations is essential. Such collaboration ensures that SEON technologies remain both technically viable and economically sustainable. Additionally, continuous research and development (R&D) efforts are crucial for addressing the evolving demands of modern optical networks, enabling scalable, efficient, and resilient SEON infrastructures. By integrating cutting-edge innovations and adhering to industry standards, SEON can achieve optimal performance, interoperability, and long-term sustainability in next-generation optical communications.

4.6. Disaster management

Disaster management [242–246] in SEON involves the development and implementation of strategies to ensure the network's ability to withstand and maintain critical communications during natural disasters, emergencies, or large-scale network failures. SEON play a vital role in disaster response and recovery by providing a reliable and adaptive communication infrastructure, ensuring the continuity of essential services when conventional networks may be compromised.

The need for restoring network resources after major natural disasters, such as tsunamis, hurricanes, and floods, is becoming increasingly critical. Unlike localized network failures such as fiber cuts or node malfunctions, disaster recovery poses unique challenges due to its unpredictable and widespread impact. While network failures can be mitigated by allocating redundant resources, planning for comprehensive disaster recovery is economically impractical given the uncertainty of catastrophic events.

A promising approach to enhancing disaster recovery in SEON is the use of Sliceable Bandwidth-Variable Transponders (SBVTs). The flexibility of SBVT technology enables dynamic resource allocation, improving network adaptability and survivability under extreme conditions [53]. As a result, research on integrating SBVTs within elastic optical networks has gained interest, particularly in the field of disaster management.

Beyond technological advancements, SEON contribute to disaster resilience by integrating strategic planning, real-time monitoring, and collaboration with emergency response services. The primary goal is to ensure the uninterrupted functionality of critical communications infrastructure during crises, ultimately supporting life-saving operations and facilitating post-disaster recovery efforts.

5. Implementation and simulation results

Survivability, particularly against single-link failures, remains a critical challenge in SEON, as these failures occur more frequently than multi-link failures and can cause severe service disruptions [6,70,75,129,175,197,198,211,247,248]. To address this, we propose a novel

protection scheme termed State-Aware Symmetrical Hybrid Protection (SHP). This approach integrates symmetrical traffic splitting and a hybrid protection mechanism grounded in the principles of SBPP. Unlike DPP, which allocates a separate backup path for each primary path (as discussed in Section 2.1), SBPP enables multiple primary paths to share a common backup path, thereby significantly reducing resource overhead and enhancing spectrum efficiency.

Several studies have explored traffic demand splitting to enhance both network survivability and spectrum utilization. For example, [249] proposed the State-Aware STD and ASTD models, which improved connection acceptance rates but did not consider failure scenarios. Similarly, the MMRSA model in [33] incorporated traffic splitting but lacked a strong survivability framework. Other works, such as [12,31], introduced PDPPS, which relies on DPP and thus misses the efficiency benefits offered by SBPP.

Our main motivation is to show that optimal resource utilization can be achieved by symmetrically splitting traffic demands, particularly when no single path can handle the entire demand. This approach also supports the spectrum management goals discussed in Section 4.2. Furthermore, we demonstrate that multiple traffic demands can be effectively protected against single-link failures without incurring additional protection overhead, highlighting the strength of SBPP based strategies as a scalable and resource efficient solution for improving SEON survivability.

This work extends the framework proposed in [249] by presenting a Hybrid Protection Scheme that integrates SBPP with a Vulnerability Scoring System (VSS). The aim is to improve survivability, maximize spectral efficiency and enable failure aware traffic engineering in SEON. Traditional SBPP enables several primary paths to share one backup path, reducing redundancy and saving resources. However, standard SBPP does not explicitly incorporate link reliability, historical outage frequency or congestion levels, which may lead to the selection of unstable or highly loaded backup paths. Such shortcomings increase blocking probability and reduce the effectiveness of recovery mechanisms. To mitigate these issues, we introduce VSS, which dynamically evaluates each link's vulnerability by accounting for slot utilization, past failure behavior and physical distance. The link score is computed as

$$V_{uv} = w_u \cdot \frac{U_{uv}}{U_{\max}} + w_f \cdot \frac{F_{uv}}{F_{\max}} + w_d \cdot \frac{l_{uv}}{L_{\max}}, \quad (1)$$

where U_{uv} is the number of occupied slots on link (u, v) , normalized by the maximum slot capacity $U_{\max}$. The variable F_{uv} denotes the historical failure count for link (u, v) , normalized by the largest observed value in the network, $F_{\max}$. The distance of each link, l_{uv} , is normalized by the longest link length in the topology, $L_{\max}$. The coefficients w_u, w_f, w_d are tunable parameters constrained by $w_u + w_f + w_d = 1$, ensuring proportional influence of utilization, reliability, and physical length. Since all three terms are bounded within $[0, 1]$, the final score V_{uv} also lies within this interval, allowing straightforward comparison across heterogeneous links.

The weight assignment $(w_u, w_f, w_d) = (0.5, 0.3, 0.2)$ reflects the relative impact of the three factors. Utilization carries the largest weight (50%) as it captures instantaneous network stress and congestion, allowing quick adaptation to traffic fluctuations. Historical failures contribute 30%, ensuring that unreliable links are penalized without disproportionately biasing against single past incidents. Distance is weighted at 20% to account for physical fragility and signal degradation without overshadowing the dynamic factors. The sum of weights is always unity, which normalizes the final score and prevents dominance by any single metric. Although we adopt a 50:30:20 split as a balanced compromise, the model is flexible: operators may increase the failure weight when reliability is paramount, or emphasize utilization for higher spectrum efficiency. In this way, VSS remains adaptable to diverse operational priorities.

Each bidirectional link in our SEON is provisioned with $C = 320$ spectrum slots, giving $U_{\max} = 320$. Failures are normalized by the

maximum recorded outage across all links, $F_{\max}$, and distances by the longest fiber span, $L_{\max}$. Thus, the per-link VSS is given by

$$V_{uv} = 0.5 \frac{U_{uv}}{F_{\max}} + 0.3 \frac{F_{uv}}{F_{\max}} + 0.2 \frac{L_{uv}}{L_{\max}},$$

and the vulnerability of a path p is computed as the average over its constituent links:

$$V_{\text{path}} = \frac{1}{|p|} \sum_{(u,v) \in p} V_{uv}.$$

Lower path scores indicate more reliable routes, guiding both primary and backup selection. By integrating SBPP with VSS in this manner, the proposed scheme proactively avoids fragile or congested links, dynamically adapts backup assignments, and ensures higher acceptance ratios with reduced blocking probability compared to conventional approaches.

The Hybrid Protection process begins by computing multiple candidate routes between a source and destination using the k -shortest path algorithm. Because k -shortest paths alone do not guarantee disjointness, we subsequently apply a link-disjoint selection procedure based on a maximum flow approach. From this, several link-disjoint primary routes and one backup route are identified. Each candidate path is ranked according to its average vulnerability score derived from VSS and the lowest-scoring primary paths are chosen for traffic allocation. The backup is then selected from the remaining disjoint options, where remaining refers to those feasible candidate paths not selected as primaries but still disjoint from them. To prioritize links, each link cost in the k -shortest path computation is defined by its vulnerability score, so that paths traversing more reliable links achieve lower costs. In this way, both primary and backup paths are selected according to vulnerability-aware metrics, ensuring that they are disjoint while also minimizing overall failure risk. Traffic is symmetrically splitted evenly across the selected primaries for balanced load sharing. Symmetrical distribution ensures that the offered demand is evenly partitioned across the selected primary paths, preventing overload on any single route. While all used paths occupy slots, asymmetrical allocation can cause one path to saturate quickly, increasing blocking probability and spectrum fragmentation, while other paths remain underutilized. By keeping the slot occupancy uniform, symmetry delays congestion, improves load balancing and provides a fairer and more resilient utilization of network resources.

The shared backup path is periodically evaluated using the VSS. If it becomes unreliable due to congestion (insufficient free spectrum on any of its links) or if its constituent links exhibit high vulnerability scores reflecting repeated historical failures, it is dynamically reassigned to a more reliable, link-disjoint candidate. In case of a failure event, affected flows are rerouted over the pre-allocated shared backup path, ensuring continuity of service. During normal operation, the backup path does not carry traffic; however, its spectrum is pre-reserved along the selected route following the SBPP model. When risk or contiguity tests fail or when the set of protected primaries changes, the controller reassigns the prereserved backup to a lower-risk, link-disjoint route and updates the amount of prereserved capacity on its links; these updates occur at admission/teardown events or periodic VSS checks, not at failover. Under the single link failure assumption, this pre-reservation guarantees instantaneous failover while allowing multiple primaries to share the same backup capacity. Consequently, the reservation overhead is lower than in DPP not because reservation is deferred (which would amount to restoration), but because SBPP reserves only the capacity required to withstand one failure at a time. In words, on each backup link we reserve an amount equal to the largest per-connection slot requirement among the primaries that share that backup, i.e., we take the maximum requirement, not the sum so any single primary failure can be absorbed while still enabling efficient sharing. If, during the admission of a new request, the candidate shared backup path lacks the required number of contiguous free slots on one or more links, we trigger a global re-optimization.

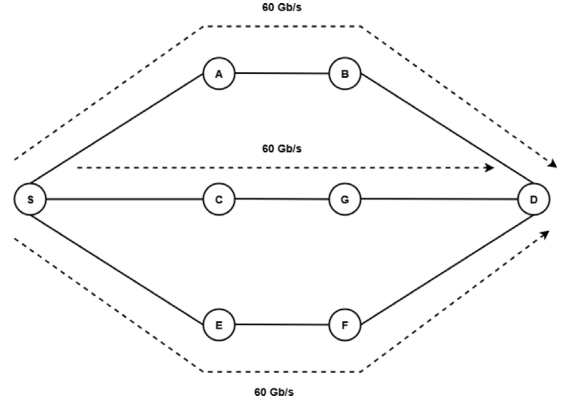


Fig. 15. PDPPS [55].

This process repacks existing allocations to free contiguous spectrum while preserving full protection for all already admitted connections. In our simulator, this is implemented by releasing and replaying current allocations to obtain a more compact packing; operationally, this corresponds to make-before-break reconfiguration without degrading service. If re-optimization cannot create sufficient contiguous capacity, the new request is blocked; existing flows remain fully backed up. During re-optimization, vulnerability scores are recomputed to reflect the updated network state. This differs from PDPPS, where traffic is simultaneously partitioned across all disjoint paths. In contrast, our model uses only the selected primaries for active transmission, while the backup remains idle until a failure occurs, as illustrated in Figs. 15 and 16.

We compare our proposed model with PDPPS since it is one of the most widely studied protection schemes in SEON. While traditional methods like DPP provide strong protection, they suffer from high resource overhead due to idle backup paths. In PDPPS, the total available spectrum on each link is partitioned into separate working and protection pools. During normal operation, working traffic occupies only the designated partition, leaving the protection partition idle but pre-reserved. Upon a single-link failure, backup paths are activated by utilizing the reserved protection partition, but instead of allocating the full demand, PDPPS employs a squeezed restoration strategy where only a fraction of the original bandwidth is guaranteed, while the remaining portion may be dropped or degraded. This partitioning mechanism enhances spectrum utilization compared to pure DPP by avoiding complete duplication of resources; however, it also introduces significant challenges. First, the rigid partitioning of spectrum can lead to fragmentation and under utilization under dynamic traffic conditions. Second, service quality may be degraded during failure scenarios since affected connections receive only partial restoration. Finally, the reliance on static partition sizes makes PDPPS less adaptive to heterogeneous demand patterns and evolving network states.

In the PDPPS [55] setup shown in Fig. 15, we pre-establish three link-disjoint lightpaths from S to D. Each path carries 60 Gb/s, so in normal operation the network provides 180 Gb/s. The offered traffic is 150 Gb/s, spread across the three paths, leaving about 30 Gb/s of headroom. Because the paths are disjoint, a single link failure can disable at most one path. If any one path fails, the two surviving paths still carry 60 Gb/s each (120 Gb/s total), so service continues without a reset. The drawback is squeezing: the missing 30 Gb/s during the outage is not served, so part of the traffic is lost as shown in the literature [250].

In contrast, our proposed State-aware SHP model eliminates these inefficiencies by dynamically allocating resources based on demand size, path distance and real-time link vulnerability. Unlike PDPPS, which reserves idle capacity upfront, State-aware SHP leverages SBPP

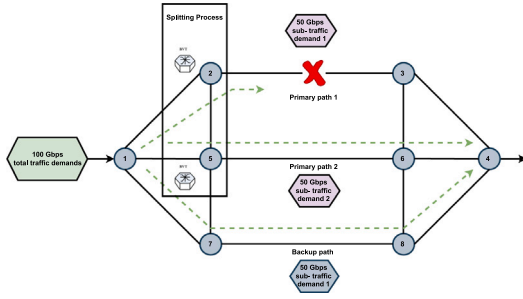


Fig. 16. State-aware SHP.

so that a single backup path can simultaneously protect multiple primary paths, thereby minimizing spectrum duplication. Furthermore, by integrating the VSS, State-aware SHP model intelligently avoids risk-prone links, ensuring higher survivability. This adaptive, state-aware design enables proposed model to achieve higher connection acceptance rates, lower blocking probability and improved spectrum utilization, while maintaining stronger service guarantees than PDPPS under both normal and failure conditions.

Note that, although the proposed model may distribute traffic across multiple paths, splitting applies only to the working traffic on link-disjoint primaries and is invoked only when no single low-risk path with the required contiguous slots is available. Protection capacity is prereserved once along a single shared backup path. This design avoids PDPPS rigid spectrum partitions and its squeezed restoration under failure. The reduction in resource duplication results from shared reservation on the backup, not from deferring reservation. Our contribution is to pair SBPP with VSS guided path ranking, symmetrical splitting across disjoint primaries when necessary and an admission-time re-optimization step tailored to SEON contiguity constraints, thereby minimizing unnecessary splitting overhead (e.g., additional guard bands, coordination, and differential-delay compensation).

As an illustrative example, the State-aware SHP model shown in Fig. 16 handles a traffic demand of 100 Gbps by symmetrically splitting it into two primary paths of disconnected links—P1: 1-2-3-4 and P2: 1-5-6-4—each carrying 50 Gbps. In the event of a failure (e.g., 2-3 link in P1), the affected traffic is immediately rerouted through the shared backup path B1: 1-7-8-4, ensuring uninterrupted service.

These model is designed to protect traffic against single link failures by accepting a connection only if both primary and shared backup paths can be successfully established. If the setup of the primary path fails, the backup path is not attempted [251–256]. This selective mechanism offers a moderate level of protection [257], reflecting a non-joint SBPP approach [258], where primary and backup paths are computed independently. This decoupled strategy simplifies the operational complexity while still maintaining a reasonable trade-off between protection and resource utilization.

In the event of a single-link failure affecting any of the primary paths, the impacted traffic is immediately rerouted to the pre-reserved shared backup path. This fast restoration process eliminates the need for real time path computation, thereby reducing downtime. To maintain long term efficiency and adaptability, the system performs periodic re-optimization. This process involves recalculating all vulnerability scores based on updated network metrics, defragmenting links where necessary and reassigning both primary and backup paths to maintain optimal operating conditions.

By addressing these scenarios, the proposed model provide a robust framework that strengthens network resilience and ensures efficient recovery in the face of failures, while optimizing spectrum use through traffic splitting, and shared-backup protection. We assess these benefits using the following metrics: network resilience, defined as the fraction of active connections that remain in service under single-link fault

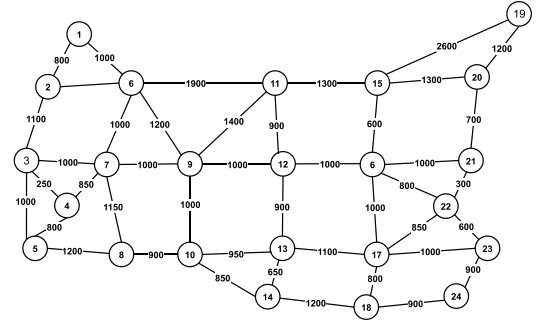


Fig. 17. USNET topology [130].

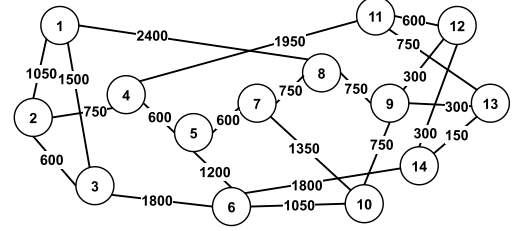


Fig. 18. NSFNET topology [259].

injection; failure recovery, defined as the ratio of successful failovers to all failover events affecting active connections (and, when reported, a service-continuity ratio given by the carried traffic immediately after failover relative to just before for the affected flows); and spectrum efficiency, measured via time-averaged spectrum utilization together with blocking probability (lower is better), with the share of admissions rejected for lack of contiguous slots reported as a qualitative indicator of fragmentation.

To validate our model, we ran simulations on the USNET and NSF NET topologies (Figs. 17 and 18) with the objectives of minimizing traffic loss, reducing blocking probability and maximizing time-averaged spectrum utilization. Connection requests follow a Poisson arrival process with rate set by the offered load in Erlangs (mean holding time = 1). Single-link failures are generated by thinning the arrival stream: at each arrival, a failure is triggered independently with probability $p_f = 0.005$, yielding a Poisson failure rate proportional to the load; when a failure occurs, one link is chosen uniformly at random, taken down for a short repair interval (0.1 time units), its failure count is recorded for VSS and the link is restored. Metrics are computed per load: spectrum utilization is the time average of the instantaneous network utilization (the average, over all links, of the occupied-slot fraction); blocking probability is the ratio of blocked to attempted connection requests (a request is blocked if continuity/contiguity cannot be satisfied on the selected primaries and/or on the pre-reserved shared backup after the admission-time defragmentation attempt); and traffic loss is total blocked demand (Gb/s) divided by total offered demand (Gb/s).

Based on the overall analysis, the State-aware SHP model consistently outperforms PDPPS across multiple metrics. For blocking probability (Blocking probability is defined in Methods as blocked/attempted requests), State-aware SHP yields a 97% reduction on USNET (Fig. 19) and an 80% reduction on NSFNET (Fig. 22). These percentages are computed over the entire operating range by integrating the blocking-versus-load curves (trapezoidal area under the curve, AUC, over $L \in [25, 200]$ Erlangs) and reporting $100 \times (1 - \text{AUC}_{\text{State-aware SHP}} / \text{AUC}_{\text{PDPPS}})$, as defined in Methods. The gain stems from VSS-guided selection of low-risk, link-disjoint primaries, symmetric splitting when a single feasible path is unavailable, and an admission-time spectrum re-optimization step that mitigates fragmentation. We adopt SBPP, a single pre-reserved backup protecting multiple link-disjoint primaries

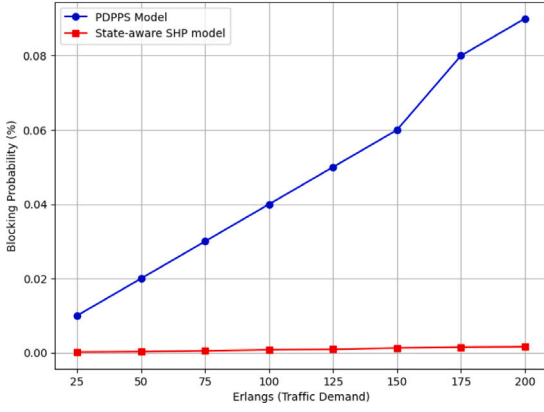


Fig. 19. Comparison of blocking probability using USNET topology.

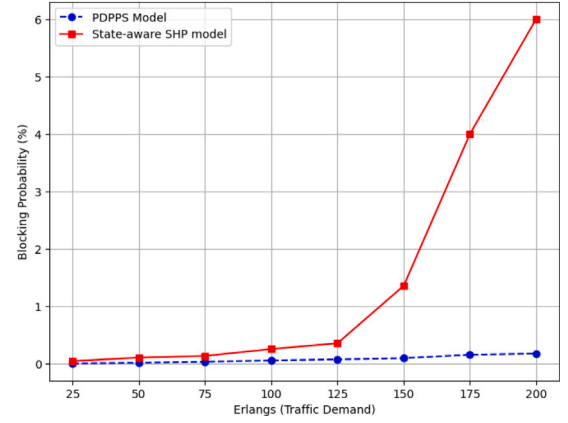


Fig. 22. Comparison of blocking probability using NSFNET topology.

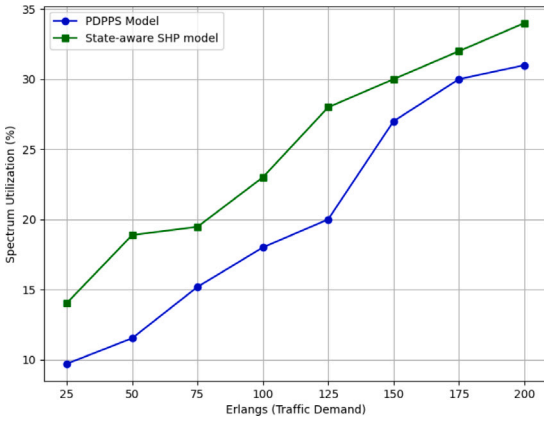


Fig. 20. Comparison of spectrum utilization using USNET topology.

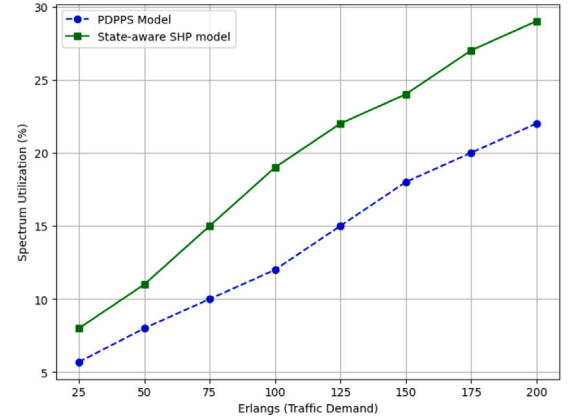


Fig. 23. Comparison of spectrum utilization using NSFNET topology.

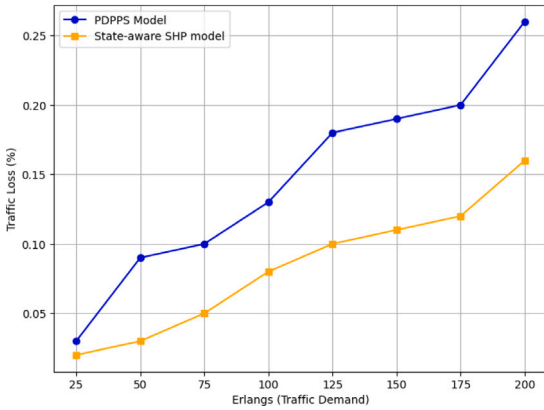


Fig. 21. Comparison of traffic loss using USNET topology.

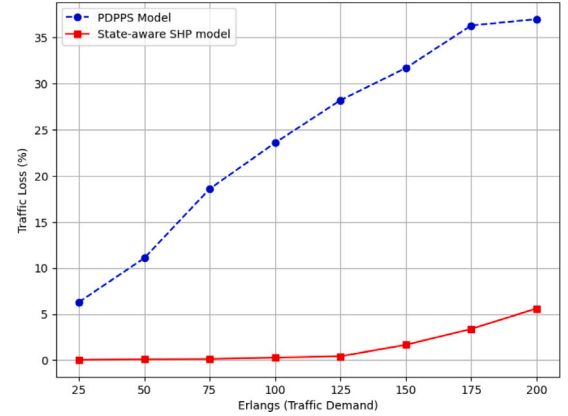


Fig. 24. Comparison of traffic loss using NSFNET topology.

under the maximum (not sum) reservation rule—and do not introduce a new protection primitive; our novelty lies in VSS guided path ranking, conditional symmetric splitting, and admission-time spectrum re-optimization tailored to EON contiguity constraints.

State-aware SHP also improves spectrum efficiency and carried traffic. The utilization of the time-averaged spectrum (defined as the time average of instantaneous network utilization, that is, the average over all links in the occupied slot fraction) increases by 29% in USNET

(Fig. 20) and 40% on NSFNET (Fig. 23), while the loss of traffic (blocked demand divided by the offered demand) decreases by 44% in USNET (Fig. 21) and 90% on NSFNET (Fig. 24). Percentages for utilization (increase) and loss (reduction) are obtained by integrating each metric versus load curve over $L \in [25, 200]$ using the same trapezoidal aggregation.

These results confirm that the State-aware SHP model, by combining traffic splitting, dynamic allocation and hybrid SBPP-based protection, offers significant improvements in spectrum efficiency, failure recovery, and network resilience. Compared to traditional approaches like DPP, which incur high resource overhead, State-aware SHP and SBPP

based strategies provide a scalable, cost-effective and robust solution for future SEON deployments, highlighting their potential to meet growing traffic and reliability demands.

6. Conclusion

This study presents a comprehensive survey of SEON, analyzing a wide spectrum of strategies and technologies designed to enhance network survivability. The techniques reviewed – including Dedicated and Shared Protection Schemes, Path and SR, Segment-based Strategies, and Hybrid Models – have been carefully selected through an in-depth literature review and current domain expertise. A detailed investigation into failure scenarios, such as single-link, multi-link, node, and hardware failures, highlights the need for adaptive and intelligent survivability mechanisms.

We have also identified critical future challenges and research directions, such as centralized network control, spectrum fragmentation management, AI/ML-driven network intelligence, PLI, hardware adaptability, and disaster resilience. Beyond current methodologies, this paper underscores the growing significance of BVTs, traffic splitting models, and real-time decision-making algorithms in optimizing spectrum utilization, fault recovery, and service continuity.

Importantly, as SEON continues to evolve, it is influenced by hardware and software advancements that are shaping the future of optical networking. Technologies such as SDM, holographic optical communications, and quantum computing-based RSA are emerging as transformative solutions that promise higher scalability, spectral efficiency, and fault tolerance. These innovations have the potential to radically expand the capabilities of SEON, enabling it to meet the ever-growing demands of next-generation applications and critical infrastructure.

In conclusion, this study not only consolidates the state of survivability in SEON but also lays the groundwork for future innovation. By bridging foundational techniques with cutting-edge research trends, we aim to inspire the development of resilient, intelligent, and adaptive SEON architectures that are better equipped to serve the increasingly dynamic and data-driven needs of our society.

CRediT authorship contribution statement

Amit Khanjan Sarma: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sanjib K. Deka:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Nityananda Sarma:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Amit Khanjan Sarma reports financial support was provided by Ministry of Electronics and Information Technology. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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